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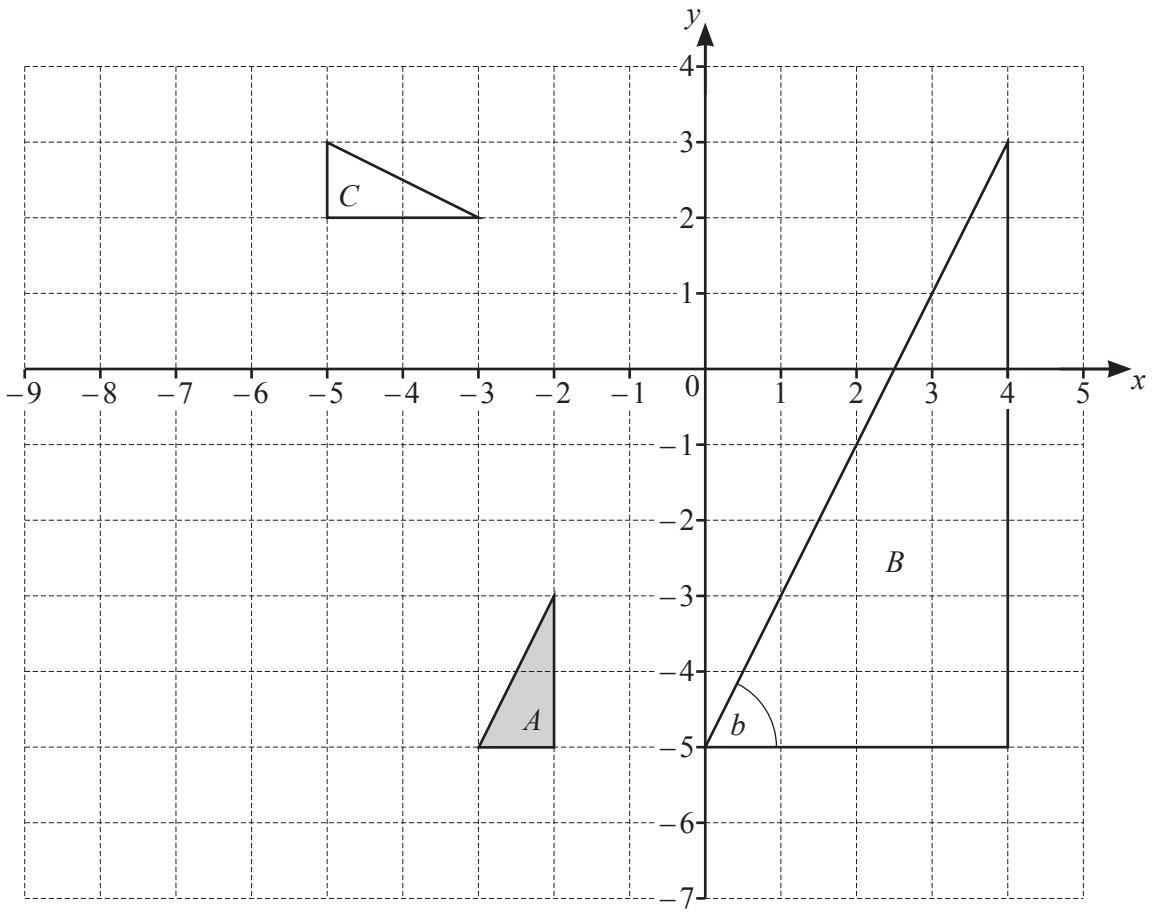
6 The diagram shows three triangles, A, B and C, on a 1 cm^2 grid.



(a) Describe fully the **single** transformation that maps

(i) triangle A onto triangle B,

.....

..... [3]

(ii) triangle A onto triangle C.

.....

..... [3]

(b) On the grid, draw the image of

(i) triangle A after a translation by the vector $\begin{pmatrix} -5 \\ 4 \end{pmatrix}$, [2]

(ii) triangle A after a reflection in the line $x = -4.5$. [2]

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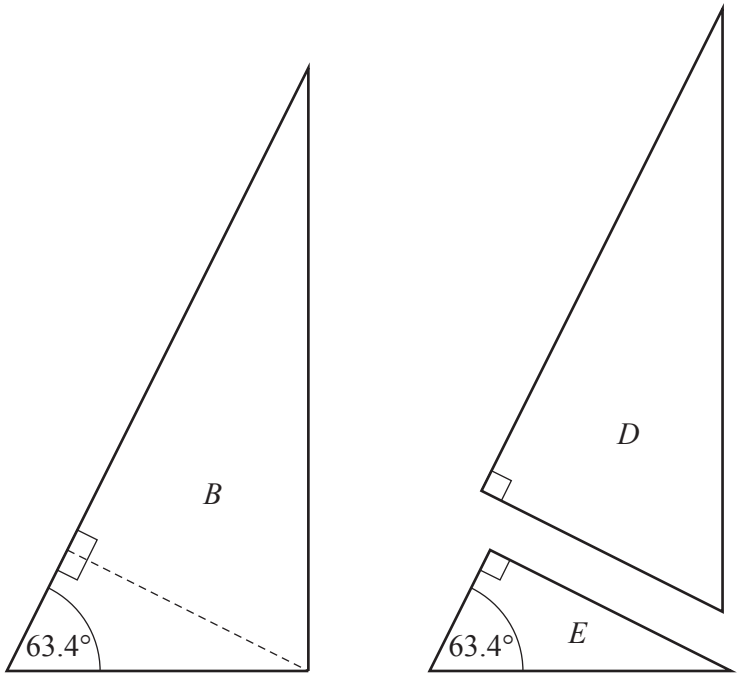
(c) The diagram also shows an angle b in triangle B.

Use trigonometry to show that angle b is 63.4°, correct to 1 decimal place.

[2]

(d)



Two new triangles, D and E, are made from triangle B, as shown in the diagram.

Are all three triangles similar?
Give a reason for your answer.

..... because .....

..... [2]